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Leveraging [^{18}F]F-Piflufolastat PET for PBPK Prediction of Tissue Distribution and Therapy Optimization in Prostate Cancer

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ABSTRACT

Prostate-specific membrane antigen (PSMA) is overexpressed in over 90% of prostate cancer tumors, making it a strong target for imaging and therapy. The diagnostic tracer [^{18}F]F-Piflufolastat provides high affinity and image quality, while the β -emitter [^{177}Lu]Lu-PSMA-617 enables targeted treatment. Despite differences in radioactive half-life, their shared PSMA-binding and pharmacokinetic profiles suggest potential for translational pharmacokinetic modeling. We hypothesized that the published full-body physiologically based pharmacokinetic (PBPK) model could be directly translated across the investigated PSMA ligands to predict tracer distribution within the same patient. A PBPK model was developed by rebuilding and modifying a published [^{68}Ga]Ga-PSMA-11/[^{177}Lu]Lu-PSMA-617 model to simulate [^{18}F]F-Piflufolastat PK. Patient-specific factors included tumor volume, PSMA receptor density, organ blood flow, and renal clearance. Verification used published data, dynamic positron emission tomography (PET) from eight patients, and static total-body PET/computed tomography (CT) at 2 h from five patients. Extrapolation to [^{177}Lu]Lu-PSMA-617, using fitted receptor density, was evaluated against 24-h and 7-day Single-Photon Emission Computed Tomography (SPECT) data from two patients using mean prediction error (PE). The model accurately predicted [^{18}F]F-Piflufolastat tumor uptake (median PE -3.44% (IQR -3.79 to -3.17) and -3.53% (IQR -3.78 to -3.00) in internal and external cohorts) and showed consistency with 7-day [^{177}Lu]Lu-PSMA-617 SPECT observations. However, predictions for other organs were variable (median PE ranging between -89.7 and 144), indicating limitations of a one-to-one molecular translational approach. These findings demonstrate that PBPK modeling can capture key determinants of PSMA tracer distribution, and link [^{18}F]F-Piflufolastat imaging to [^{177}Lu]Lu-PSMA-617 predictions. However, improved representation of systemic and organ-level kinetics is required. Future work should explore a limited PBPK approach, focusing on accurate blood kinetics and organ-specific modeling.

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Study Highlights

- What is the current knowledge on the topic?
 - PSMA-targeted PET imaging and radionuclide therapy are established in prostate cancer, using tracers such as [^{18}F]F-piflufolastat and therapeutics such as [^{177}Lu]Lu-PSMA-617. Despite their shared target, current practice relies on fixed dosing, resulting in substantial interpatient variability in tumor and organ exposure. PBPK models have been developed for PSMA ligands and identified tumor blood flow and receptor density as key determinants, but it remains unclear whether these models can be directly translated across structurally different ligands to link diagnostic imaging with therapeutic distribution.
- What question did this study address?
 - Can a previously published full-body PBPK model be directly translated to [^{18}F]F-piflufolastat and subsequently used to predict tissue distribution of [^{177}Lu]Lu-PSMA-617 within the same patient?
- What does this study add to our knowledge?
 - This study shows that PBPK modeling accurately predicts tumor uptake of [^{18}F]F-piflufolastat and allows estimation of patient-specific PSMA receptor density, enabling prediction of [^{177}Lu]Lu-PSMA-617 tumor exposure. However, predictions for blood and several organs were less accurate, demonstrating that direct one-to-one translation between PSMA ligands does not fully capture organ-specific pharmacokinetics. Simulations revealed substantial variability in predicted therapeutic doses (0.15–13.28 GBq) to achieve comparable tumor exposure.
- How might this change clinical pharmacology or translational science?
 - These findings highlight both the potential and limitations of PBPK modeling for imaging-to-therapy translation. While supporting its use as a mechanistic tool to explore interpatient variability and guide hypothesis generation for precision dosing, they also indicate that refined, possibly reduced PBPK models are needed before clinical implementation in individualized PSMA-targeted therapy.
- Question
 - Can PBPK modeling of diagnostic imaging data inform individualized therapeutic dosing for PSMA-targeted radionuclide therapy?
- Pertinent findings
 - A mechanistic PBPK model, informed by patient-specific baseline characteristics and [^{18}F]F-Piflufolastat PET imaging, adequately described tumor uptake and was used to simulate for [^{177}Lu]Lu-PSMA-617 pharmacokinetics. Simulations indicated substantial interpatient variability in predicted tumor exposure, with an estimated dose range of 0.15–13.28 GBq required to achieve

comparable tumor target attainment. However, predictive performance for several organs-at-risk was limited, and direct translation between PSMA ligands was not consistently reliable.

- Implications for patient care
 - These findings suggest that PBPK modeling may be useful as an exploratory tool to investigate variability in PSMA-targeted therapy and to generate hypotheses on individualized dosing. At present, limitations in the model performance, particularly for organs-at-risk, and uncertainty in cross-ligand translation restrict clinical applicability. Further methodological refinement and prospective validation are needed before such approaches can inform patient-specific dosing in practice.

1 | Introduction

Prostate cancer (PCa) is one of the main health concerns affecting men and is the most prevalent type of cancer among middle-aged and older men [1]. Prostate-specific membrane antigen (PSMA) is a transmembrane protein that is primarily expressed on more than 90% of PCa cells, particularly in high-grade and dedifferentiated tumors [2–4]. Due to its high expression on PCa cells and the strong affinity of PSMA radioligands, PSMA is a valuable target for diagnostic imaging and targeted radionuclide therapy.

For diagnostic purposes, the PSMA positron emission tomography (PET) tracer, [^{18}F]F-Piflufolastat (Pylclari) is a clinically established radiotracer, valued for its high affinity for PSMA receptors, convenient half-life time, and excellent image quality. Meanwhile, the β -emitting radioligand [^{177}Lu]Lu-PSMA-617 (Pluvicto) serves as a potent therapeutic radiotracer. Theranostics, which integrates diagnostic and therapeutic modalities using ligands with equal targets, plays a significant role in PSMA-targeted therapies. Although [^{18}F]F-Piflufolastat and [^{177}Lu]Lu-PSMA-617 are not a true theranostic pair due to differences in structure, pharmacokinetics, and binding characteristics, they share the same target and exhibit comparable PSMA receptor-binding profiles [5, 6]. These similarities allow for a meaningful comparison of their distribution within the same patient, despite differences in drug-specific properties such as radioactive half-life time. However, for [^{18}F]F-Piflufolastat it is suggested that it has superior tumor-to-background compared to gallium-based tracers [7]. By prioritizing the superior tumor-to-background contrast and diagnostic performance of [^{18}F]F-Piflufolastat over gallium-based tracers and combining it with the therapeutic efficacy of [^{177}Lu]Lu-PSMA-617, this approach bridges the link between high-precision imaging and targeted treatment [7]. Ultimately, the integration of their distinct pharmacological strengths optimizes both diagnostic accuracy and therapeutic efficacy.

Currently, in PCa theranostics, PSMA-based radionuclide therapy follows a standardized fixed-dose approach for imaging and therapy of 300 MBq and 7.4 GBq, respectively [8–11]. This one-dose-fits-all strategy leads to variations in exposure to target

tissues (prostate and tumor) as well as organs-at-risk (OAR) among patients. Low target exposure may result in decreased therapeutic benefit, whereas high exposure in OAR increases the risk of toxicity, such as dry mouth or xerostomia [12, 13]. When uptake of [^{18}F]F-Piflufolastat is quantified during diagnostics, dose optimization is possible to increase effectiveness by exposing the tumor to the optimal dose and to reduce toxicity by lower exposure to the OAR.

Pharmacokinetic (PK) modeling can be used to quantify drug and patient specific variability in PK. Prior modeling studies were applied to predict drug uptake of [^{68}Ga]Ga-PSMA and [^{177}Lu]Lu-PSMA radioligands. These models showed significant interpatient variability in tumor and organs-at-risk (OAR) exposure after PSMA-targeted treatments [14–17]. Tissue characteristics, such as tumor volume, receptor density, and organ blood flow, are the likely cause of this variability [8, 16–18]. Physiologically-based pharmacokinetic (PBPK) can be applied in a mechanistic approach to (1) explore differences in PK characteristics between PSMA-based radioligands and (2) identify patient and (tumor) tissue characteristics that predict the intra- and interindividual variability in drug distribution during diagnosis and treatment.

As a first step, this study aimed to investigate whether a previously published full-body PBPK model could be directly applied across the investigated PSMA ligands ([^{18}F]F-Piflufolastat and [^{177}Lu]Lu-PSMA-617). Specifically, we evaluated whether this mechanistic PBPK framework could be adapted to describe the PK of [^{18}F]F-Piflufolastat and subsequently predict tissue distribution of the therapeutic ligand [^{177}Lu]Lu-PSMA-617 within the same patient. We hypothesized that the shared PSMA-targeting mechanism of these investigated ligands would allow direct translational use of the full-body PBPK model to link diagnostic imaging with therapeutic tracer distribution.

2 | Materials and Methods

2.1 | Model Development

To develop a PBPK model, physiological parameters, such as blood flow, organs volumes, and renal elimination are included [19]. This PBPK model accounts for the heterogeneity of physiological processes among various individuals, compared to other models generally used to describe PET PK (e.g., two step approaches or population PK models). Additionally, PBPK models take into account differences in kinetics between tracer and therapeutic doses, as they account for (non-)linear parameters, such as binding affinity, and internalization rates of PSMA-targeting ligands.

2.2 | Description of the Literature-Derived Whole Body PBPK Model (Base Model)

A PBPK base model was derived from the study conducted by Begum et al., describing [^{68}Ga]Ga-PSMA-11 and [^{177}Lu]Lu-PSMA-617 modeling (Figure 1A) [16, 17]. This model encompassed relevant physiological and physical parameters, including blood flow, receptor density, differentiation between

PSMA-expressing and non-PSMA-expressing organs, internalization of radioligand, and excretion [16]. For each tissue, compartments are classified as either PSMA-expressing or non-PSMA-expressing organs (Figure 1A). The model describing PSMA-expressing organs consists of differential equations to describe receptor binding on the cell surface, including their binding affinities (K_{on} and K_{off}), and the internalization of the receptor-ligand complex, taken from literature (Figure 1A, Table 1).

While the model parameters are primarily sourced from Begum et al. (see [Supporting Information: Part B](#)) for PSMA-11 and PSMA-617, drug-specific adaptations for each drug were applied, including adaptations in the kidney compartment and receptor binding [16].

2.2.1 | Adaption in Kidney Compartment

In order to replicate excretion of the PSMA-ligands, the kidney compartment of the PBPK base model was adapted. A physiological model was utilized, where estimated glomerular filtration rate (eGFR) was employed to describe the primarily renal excretion of unbound [^{18}F]F-Piflufolastat [22]. The renal clearance of all compounds was based on glomerular filtration of the unbound drug, with slight differences in reabsorption and filtration accounted for by the sieving coefficient. The sieving coefficient, defined as the ratio of the concentration of a substance in the filtrate to that in plasma, was used to quantify the efficiency of glomerular filtration for the unbound radioligand.

To estimate the total amount of PSMA ligand in the vascular space over time, factors such as kidney flow and the amount filtered by the nephrons were considered. Within the kidney lumen, the presence of PSMA receptors led to a reduction in unbound ligand, as it can bind to these receptors and subsequently undergo internalization into the proximal tubules. For a more detailed depiction of the kidney compartment and its relation to kidney physiology, please refer to Figure 1B.

2.2.2 | Addition of Receptor Binding Model

To describe receptor affinity, the model utilized the receptor model described by Winter et al., as a basis [26]. Two compartments were employed to describe the PSMA binding in PSMA-expressing organs. The two compartments aimed to capture the changes in the amount of available free PSMA receptors over time. Association and dissociation rates were considered similar for [^{18}F]F-Piflufolastat and [^{177}Lu]Lu-PSMA-617 (Table 1). The receptor model determines the amount of receptor-bound ligand (AR_i) based on an inverse correlation. The baseline amount of receptors (RF_i), which are available for binding, was calculated based on the volume (V_i) and receptor density ($R_{\text{dens},i}$) of the specific organ. After the bolus injection of [^{18}F]F-Piflufolastat or [^{177}Lu]Lu-PSMA-617, the amount of free receptors may decrease due to the association of the radioligand to the receptor and subsequently internalization of the receptor-ligand complex into the cell. However, the amount of free PSMA receptors can also increase through dissociation of radioligand from the receptor (see Equation S14 and S15).

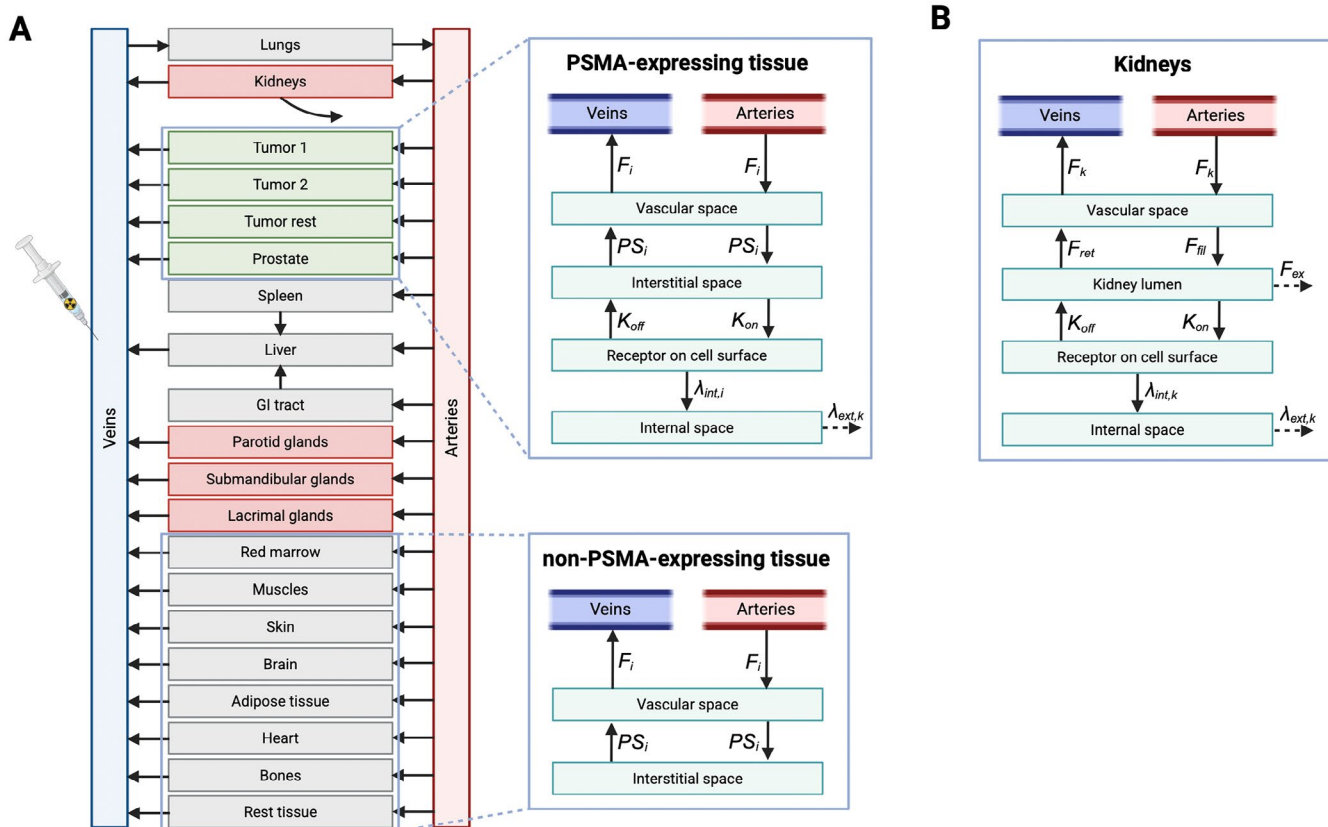


FIGURE 1 | (A) Schematic representation of whole body PBPK model (left). The model consists of various compartments representing different organs, veins, and arteries. Target tissues are depicted in green, organs-at-risk are shown in red, and non-PSMA-expressing organs and tissues are represented in gray. The subcompartment distinguishment between PSMA-expressing and non-PSMA-expressing tissues is shown on the right. The flow in and out of the vascular space is described by the organ flow F_i . The movement between the vascular and interstitial space is described by the permeability surface product PS_i . For PSMA-expressing compartments is the association to and dissociation from the PSMA-receptors on cell surface described by the k_{on} and k_{off} . The internalization by $\lambda_{int,i}$. (B) Schematic representation of kidney compartment of PBPK model. The organ flow in and out of the vascular compartment is described by F_i . The filtration flow of the kidney is described by F_{fil} and excretion by F_{ex} . The association to and dissociation from the PSMA-receptors on cell surface are described by the k_{on} and k_{off} . The internalization is described by $\lambda_{int,i}$.

Because PSMA receptors are subject to stability and recycling processes, in the receptor binding model, two assumptions were made for the receptor stability and a fast recycling process. The stability of the receptors is important because of its influence on the pharmacokinetics and -dynamics, potentially leading to variations in distribution and excretion. Regarding the recycling process of PSMA receptors, firstly, it was assumed that there is no physiological synthesis or degradation of PSMA receptors. Secondly, the model assumed a recycling process of the receptor after internalization and release of the radioligand inside the cell, without any delay in time.

For model development and verification, the original model developed by Begum et al. (see [Supporting Information: Part B](#)) was internally and externally verified against a new set of 8 and 5 patients, respectively, treated in Amsterdam UMC, and for the external cohort to a single timepoint at 120 min. Finally, the model was verified for extrapolation to [^{177}Lu]Lu-PSMA-617 using two new patients treated with their first cycle of [^{177}Lu]Lu-PSMA-617.

In addition, we adjusted for radioactive decay to injection time. The calculated total amount of [^{18}F]F-Piflufolastat and [^{177}Lu]

Lu-PSMA-617 was used as input of the administered dose for the model, and no further decay correction for [^{18}F] or [^{177}Lu] was needed in the modeling process.

2.3 | Predictive Covariates

Individually measured tumor volumes, age-independent blood flow to kidney, eGFR on kidney filtration and bodyweight on volume of distribution of the venous pool were included in the model (Table S4) based on prior research results [27–36]. As in the base model described by Begum et al., patient-specific PSMA receptor density of the primary tumor was estimated using an iterative fitting approach based on imaging data [11], for details see [Supporting Information: Part B](#). The final estimated tumor receptor density for each patient was then used to predict the full PK profile across all dynamic time points.

2.4 | Patient Data Quantification

Extraction of the patient data for this study was approved by the Human Ethics Committees of Amsterdam UMC. The study

TABLE 1 | Drug specific parameters used for whole-body PBPK model.

Parameter	Definition	[¹⁸ F]F-Piflufolastat	[⁶⁸ Ga] Ga-PSMA-11	[¹⁷⁷ Lu] Lu-PSMA-617	Unit	References
k_{on}	Association rate	0.046	0.1	0.1	L/nmol/min	[20, 21]
K_d	Dissociation constant	0.49	3.13	3.3	nmol/L	[21]
	Sieving of tracer	0.75	0.75	0.66	Ratio	[11, 22]
Fra _{ex}	Excreted fraction of ligand	0.96	—	0.67 ± 0.18	Fraction	[20, 23, 24]
Dose	Molar dosage of ligand	A _{inj} /A _{spec}	A _{inj} /A _{spec}	A _{inj} /A _{spec}	Nmol	[20, 25]
A _{inj}	Mean activity of administered	0.30	0.185	7.4	GBq	[20, 25]
A _{spec}	Specific activity	0.045	0.142	Not determined	GBq/nmol	[20, 25]
T _{1/2}	Half-life time	110 min	68 min	6.7 days		

was performed following ethical standards set forth by the Declaration of Helsinki. All patients provided written informed consent before enrollment in the study. Three cohorts of patients were used for the verification and accuracy check of the predictions of tissue distribution of [¹⁸F]F-Piflufolastat and [¹⁷⁷Lu]Lu-PSMA-617 (see [Supporting Information](#): Part B for details on patient cohorts).

2.5 | Model Verification

The accuracy of the mechanistic PBPK model was assessed using the mean prediction error (PE), calculated using Equation A. The model was considered accurate if the mean predicted concentrations fell within a 20% margin for tumors and a two-fold range for OAR of the observed patient concentrations, in line with general guidelines for model verification that define an acceptable two-fold range [37–40].

$$PE = \left(\frac{\text{Pred} - \text{Obs}}{\text{mean}(\text{Pred and Obs})} \right) \times 100 \quad (\text{A})$$

The first step of predicting the tumor uptake of [¹⁸F]F-Piflufolastat was verified against actual patient data from the three above described cohorts. By evaluating the model's performance against the observed data, this verification approach helps to determine the ability to accurately predict the tissue distribution of [¹⁸F]F-Piflufolastat in future diagnosed PCa patients and identify relevant patient characteristics. To assess the impact of individual model parameters on the predicted tissue distribution of [¹⁸F]F-Piflufolastat and [¹⁷⁷Lu]Lu-PSMA-617, a local sensitivity analysis was performed (for details see [Supporting Information](#): Part B).

2.6 | Precision Dosing Simulations

Using the gathered data of baseline characteristics and tumor predictions, we used the [¹⁷⁷Lu]Lu-PSMA-617-adjusted PBPK model to simulate the median and 95% prediction intervals of the PK profiles after the standard therapeutic dose of 7.4 GBq [¹⁷⁷Lu]Lu-PSMA. Then, we estimated the predicted dose for

each patient in which the uptake at Day 7 in the primary tumor was normalized to the median, to provide an estimate of the dose ranges that would be achieved if all patients were to receive the same tumor target attainment.

2.7 | Software

Simulations were performed using R and RStudio and packages deSolve and ggplot2 (version 4.0.3; R Foundation for Statistical Computing, Vienna Austria). Data transformation and modeling was conducted using the dplyr (version 1.1.4), tidyverse (version 2.0.0), and minpack.lm (version 1.2.4) packages. The model is openly available on GitHub at <https://github.com/ibartelink/Target-Site-PBPK-in-oncology/tree/main>. All model parameters used in the model are described in Table S1.

3 | Results

3.1 | Patients

Baseline patient characteristics of all patients are shown in Table 2.

3.2 | Model Verification

Internal verification results from eight metastatic PCa patients demonstrated the model's precision in predicting the concentration time profiles of [¹⁸F]F-Piflufolastat in tumor, only not yet within arterial blood, and lungs. Concentration time profiles for each patient and tissue for the eight dynamically scanned patients from 0 to 2 h after dosing are presented in Figure 2A–H [20, 24]. These plots show that in the rebuild model, blood activity concentrations were adequately predicted at early time points, but were overpredicted at the 120 min timepoint in all patients (PE median 112, IQR 101–115, Table S2). The kidneys showed an accurate prediction (PE median 7.13, IQR –2.47 to 82.5, Table S2), where the parotid glands and non-PSMA expressing organs such as lungs and spleen were overpredicted in the diagnostic simulations, with a median PE of 116, 119, and

TABLE 2 | Patient baseline characteristics.

Verification cohort	Patient	Age (years)	Body height (cm)	Body weight (kg)	PSA (ng/mL)	Tracer dose (MBq)	Measured tumor volume (cm ³)			
							Tumor lesion 1	Tumor lesion 2	Tumor REST	Total tumor volume
Internal	1	67	171	100	41.7	319.4	5.18	3.58	10.82	19.58
	2	63	191	119	473	301.9	6.66	6.53	19.20	32.39
	3	68	187	80	1.7	303.8	6.85	2.62	1.73	11.20
	4	65	181	81	568.4	283.8	14.40	9.86	14.46	38.72
	5	71	177	64	2792	303.6	27.58	21.38	69.31	118.27
	6	81	173	98	520	245.3	6.27	6.21	26.24	38.72
	7	71	181	85	925.2	292.8	5.31	4.54	11.58	21.43
	8	56	186	104	39.1	317.1	5.82	3.01	3.07	11.90
	Median	67.5	181	91.5	496.5	302.8	6.47	5.38	13.02	26.91
	IQR	65–71	176–186	81–101	41.1–657.6	290.6–307.1	5.69–8.74	3.44–7.36	8.88–21.0	17.66–38.72
External	9	78	178	72	2.6	313.5	ND	ND	ND	38.29 ^a
	10	82	184	90	32	317.0	ND	ND	ND	36.55 ^a
	11	75	181	108	138	308.4	ND	ND	ND	63.12 ^a
	12	86	184	77	60	314.5	ND	ND	ND	35.83 ^a
	13	73	179	81	9.3	319.1	ND	ND	ND	4.36 ^a
Therapeutic	Median	78	181	81	32	314.5	ND	ND	ND	36.55
	IQR	75–82	179–184	77–90	9.3–60	313.5–317	ND	ND	ND	35.83–38.29
	14	72	182	82	2.3	279.4	1.98	1.69	1.82	5.49
	15	75	170	90	19.6	285.3	8.16	NA	NA	8.16

Note: Patient 1–8 are derived from Janssen et al. and patient 9–13 are recent diagnostic data available from total body [¹⁷⁷Lu]Lu-PSMA-617 scans were available. Abbreviations: NA = not applicable, ND = not determined, PSA = prostate-specific antigen, SD = standard deviation. ^aOnly total tumor volume measured.

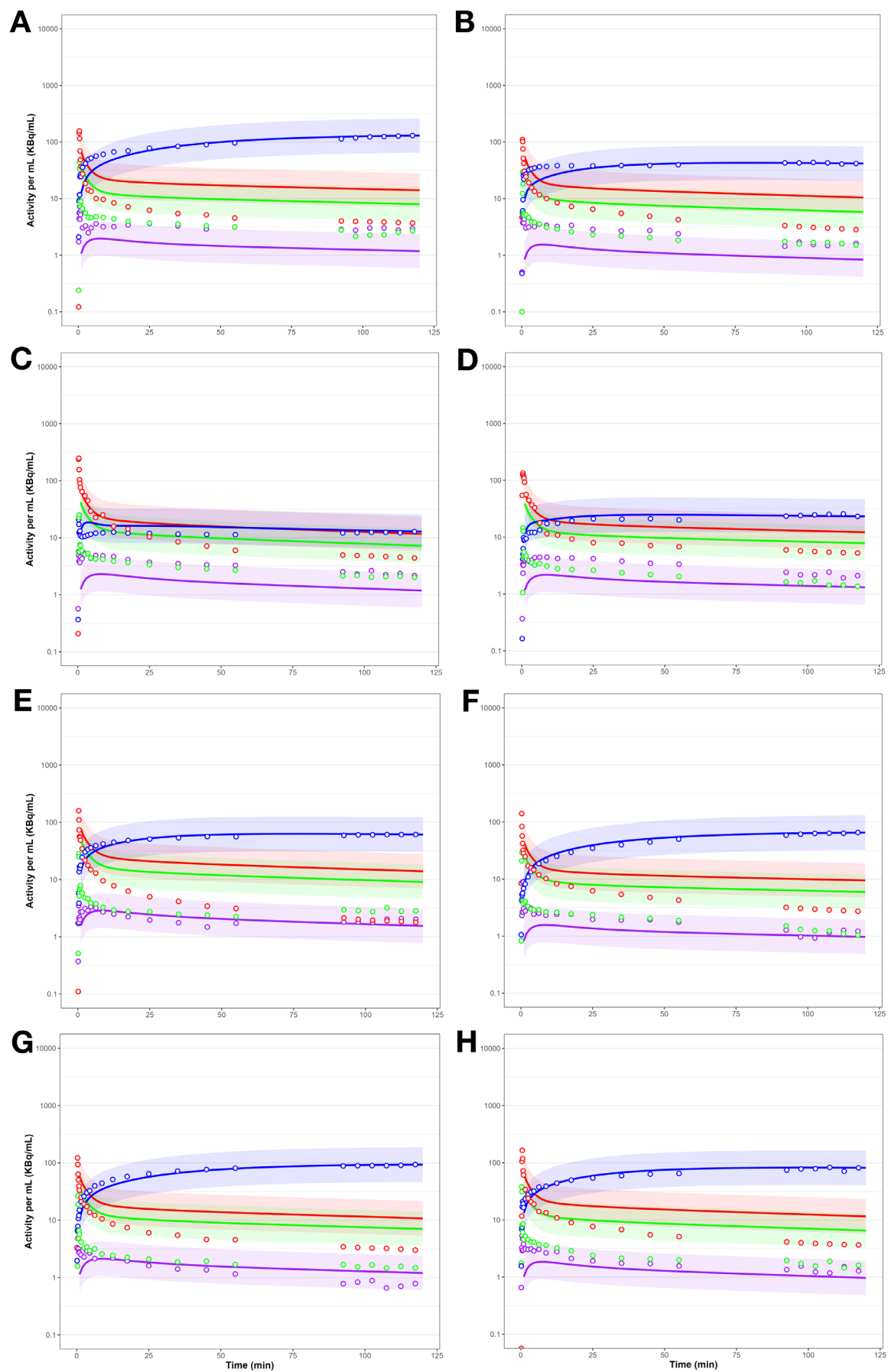


FIGURE 2 | Legend on next page.

FIGURE 2 | (A–H) [^{18}F]F-Piflufolostat tissue distribution over time in primary tumor tissue (blue), arterial blood (red), lungs (green), and muscles (purple) in a eight patients from the internal verification cohort with prostate cancer metastases. The model prediction is represented in the solid line, with corresponding observations in points and the two-fold interval in a ribbon.

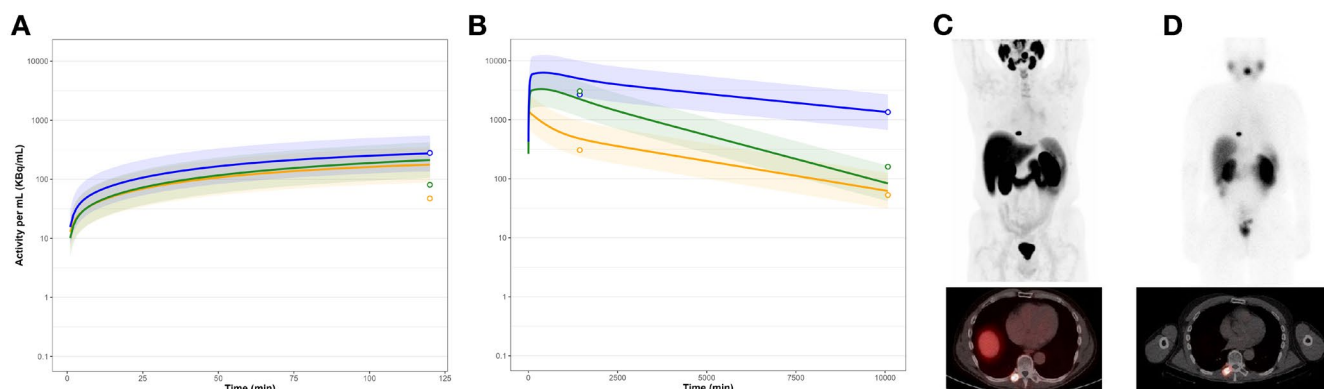


FIGURE 3 | Diagnostic prediction and therapeutic PK-simulation. (A) Concentration-time-curve of tumor (blue), kidney (orange), and parotid (green) tissue quantified during 2 h after diagnostic [^{18}F]F-Piflufolostat scan (patient 14). These Individual patient-specific parameters from the diagnostic scan and baseline characteristics were then used for the prediction. The model prediction is represented in the solid line, with corresponding observations in points and two-fold interval in a ribbon. (B) Concentration-time-curve of tumor (blue), kidney (orange) and parotid (green) tissue quantified after [^{177}Lu]Lu-PSMA-617 administration. PET-CT imaging for the same patient as presented (C) and SPECT-CT imaging at 24 (D) post [^{177}Lu]Lu-PSMA administration.

162, respectively. Other large non-PSMA expressing organs liver and muscle were underestimated with a median PE of 37.0 and 89.7, respectively.

The median prediction error in tumor tissue was -3.44% (IQR -3.79 to -3.17) in the internal verification dataset and -3.53% (IQR -3.78 to -3.00) in the external verification set. Individual prediction errors for each patient and tissue are presented in Table S3. While tumor predictions were accurate, the direct translational approach of the full-body PBPK model did not adequately predict concentrations in several organs, including lungs, kidneys, liver, spleen, and parotid glands. This indicates that a direct molecular translation of the PBPK framework by Begum et al. to other PSMA tracers does not fully capture organ-specific differences in tracer PK.

Using simulations in R, predictions were personalized based on individual patient characteristics, image-based PSMA receptor density estimation using the [^{18}F]F-Piflufolostat PET/CT scan (Table S2). Individual tumor receptor densities were subsequently applied to predict the PK profile of [^{177}Lu]Lu-PSMA in these two patients and compared with the semi-quantified SPECT data at 1 and 7 days, ensuring accuracy in predicting its distribution over time. Predictions for tumor uptake were within a two-fold prediction error, whereas predictions for blood and parotid glands were outside this range, similar to the observations for the diagnostic tracer (Figure 3, Table 2).

Sensitivity analysis indicated that tumor blood flow exerted the greatest influence on predicted tracer uptake, followed by receptor density and tumor flow density (Figure 4A). As the model was fitted to the 120 min PET timepoint, flow density was fixed at $0.135\text{ mL/min} \cdot \text{g}$, and tumor receptor densities were fitted. Among physiological parameters, eGFR and body

weight showed moderate sensitivity, whereas body mass index and hematocrit had minor effects. Other parameters had negligible impact. Overall, tumor blood flow and receptor density were identified as the key determinants of [^{18}F]F-Piflufolostat distribution, emphasizing their significance in PBPK modeling for PSMA-targeted imaging and therapy.

3.3 | Precision Dosing Simulations

Using the data of baseline characteristics and [^{18}F]F-Piflufolostat-based tumor predictions of Table 2 and S3, we simulated the median and 95% prediction intervals of the PK profiles in all 15 patients after therapeutic doses of [^{177}Lu]Lu-PSMA. The predicted range in the predicted doses was high: $0.15\text{--}13.28\text{ GBq}$, if all patients were to receive the same tumor target attainment, depicted in a distribution plot in Figure 4B.

4 | Discussion

In this study, we evaluated whether a previously published full-body PBPK model could be directly translated across the investigated PSMA-targeting ligands to link diagnostic imaging with therapeutic tracer distribution within the same patients. Current PSMA-based radionuclide therapies, such as [^{177}Lu]Lu-PSMA-617, follow a fixed-dose approach, which can lead to substantial variability in target and OAR exposure between patients [8–11]. The model accurately described tumor uptake of [^{18}F]F-Piflufolostat and enabled estimation of patient-specific PSMA receptor density, which was subsequently used to predict the pharmacokinetics of [^{177}Lu]Lu-PSMA-617. Tumor predictions were generally within $\sim 20\%$ of observed values across both diagnostic and therapeutic scans, indicating that the

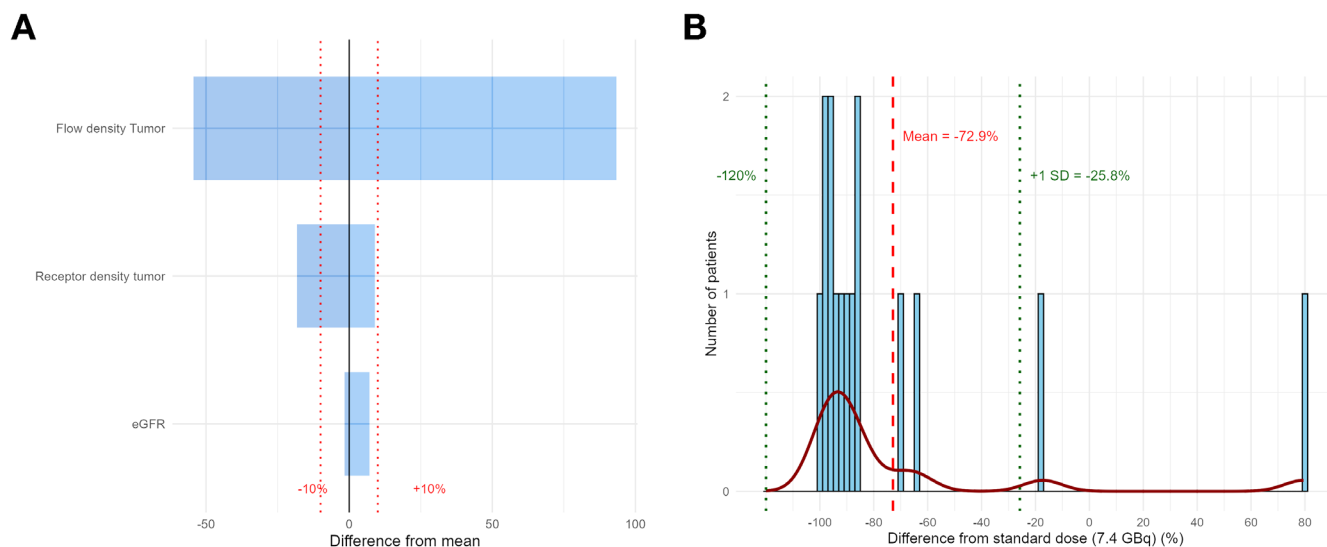


FIGURE 4 | (A) The influence of the covariates on PSMA radioligand distribution to tumor tissue. A tornado plot for the 120 min designated time point after [^{18}F]-Piiflufolastat dosing. The dotted lines mark the 10% difference from the mean, as indicator for clinical relevance. Parameters not displayed in the plot showed deviations of $<5\%$. (B) Distribution plot showing the distribution of patients according to the percentage difference from the standard administered dose of [^{177}Lu]-Lu-PSMA (7.4 GBq). The red dashed line represents the mean deviation, while the green dashed line indicates one standard deviation above the mean. The blue bars depict the number of patients within each interval, and the overlaid red curve represents the density estimate of the distribution.

model captures key determinants of tumor uptake. This performance is likely driven by the patient-specific estimation of tumor receptor density, which emerged as a major source of interindividual variability, consistent with previous studies [41–43]. These findings support the notion that population-average parameters are insufficient to describe tumor uptake and that individualized parameter estimation is required. In addition, heterogeneity between tumor lesions within the same patient may further contribute to variability in tracer distribution. Differences in vascularization, viable tumor fraction, and local PSMA expression can result in lesion-specific kinetics that are not fully captured when tumors are represented as a single homogeneous compartment, suggesting that future model extensions may benefit from incorporating lesion-level heterogeneity [44–49].

Despite the accurate prediction in tumor uptake, the model did not consistently reproduce tracer concentrations in blood and other organs. Blood concentration were well described during the early post-injection phase but were systematically overpredicted during the distribution and elimination phases. Because blood serves as the central distribution compartment, inaccuracies at this level directly affect predicted concentrations in both tumor and OAR. A consistent pattern was observed across tissues, with adequate predictions for PSMA-expressing organs such as the kidneys, overprediction in several smaller or moderately perfused organs (e.g., lung and spleen), and underprediction in large, high-capacity tissues such as liver and muscle. This balance can be interpreted from a mass balance perspective. In a mechanistic PBPK framework, total tracer mass should be conserved across all compartments. Underprediction of uptake in large organs, particularly muscle, which represents a substantial fraction of total body mass, implies that too little tracer is distributed to these tissues in the model. As a result, an apparent excess of tracer remains in the central compartment,

leading to overprediction of blood concentration at later time points. Even relatively small inaccuracies in distribution parameters for such large tissues can therefore have a disproportionate impact on systemic concentrations. The observed imbalance suggests that the current model does not fully capture the redistribution of tracer between blood, tumors, and high-capacity tissues when translated between ligands. This may reflect incomplete representation of tissue-specific uptake processes, underestimation of non-specific binding, or inaccuracies in clearance pathways. For example, renal elimination may be underestimated due to missing active transport or tubular secretion mechanisms, while tracer retention in liver and muscle may be higher than currently assumed. In addition, heterogeneity across multipel tumor lesions may further contribute to this imbalance if total tumor uptake is not fully captured by the compartment representation.

These findings indicate that direct one-to-one molecular translation of the full-body PBPK model between structurally different PSMA ligands may not adequately capture organ-specific PK processes. A logical next step is therefore the development of a limited PBPK framework, in which systemic PK are first characterized by accurately estimating blood concentration-time profiles, followed by targeted estimation of tissue parameters for clinically relevant compartments such as tumor and OAR. By reducing model complexity and focusing on key tissues, such an approach may improve parameters identifiability and predictive performance while retaining the mechanistic advantages of PBPK modeling.

Beyond predicting tracer distribution, PBPK models provide a framework to investigate treatment response and dynamic biological processes. Future extensions should incorporate more detailed descriptions of receptor turnover, including synthesis, degradation, and realistic representations of receptor

internalization and recycling, which are currently simplified [26]. In addition, treatment-induced changes in tumor burden and PSMA expression are not included but may significantly affect tracer kinetics during repeated therapy cycles [50]. Incorporation of these processes into a combined PBPK-pharmacokinetic framework may improve the prediction of longitudinal treatment outcomes.

While the PBPK offers mechanistic insight into tracer kinetics, population PK (popPK) modeling provides a complementary data-driven framework to quantify interindividual variability and identify covariates influencing exposure. PopPK models improve fit and enable evaluation of variability sources such as body composition, renal function, or tumor burden, supporting precision dosing [51]. However, their empirical nature limits mechanistic interpretation compared to PBPK models. Integrating both approaches—using PBPK-derived parameters within a population framework—may combine mechanistic insight with predictive flexibility, advancing individualized therapy for PSMA-targeted radioligands.

A major strength of this study is the use of dynamic PET/CT data for both model development and validation, enabling evaluation of early tracer kinetics as well as later therapeutic distribution in tumor and selected organs. This provides a first step toward the integration of mechanistic PK modeling with clinical PSMA theranostics.

5 | Conclusion

In this study, we evaluated whether a previously published full-body PBPK model could be directly translated across the investigated PSMA ligands to link diagnostic imaging with therapeutic distribution. While the model adequately described tumor uptake of [^{18}F]F-Piflufolostat, predictions for several organs remained suboptimal, indicating limitations of direct molecular translation between PSMA tracers. These findings suggest that further refinement of the modeling framework is needed. A limited PBPK approach, focusing on systemic pharmacokinetics and tissue penetration in tumors and organs-at-risk, may provide a more robust strategy for translating diagnostic imaging into individualized PSMA-targeted radionuclide therapy.

Author Contributions

All authors wrote the manuscript. Suzanne van der Gaag, Habibe Yilmaz, Imke H. Bartelink, Daniela E. Oprea-Lager, and Harry Hendrikse designed the research. Suzanne van der Gaag, Habibe Yilmaz, Jan Amsu, Sebastian J.V. Pruijn, and Imke H. Bartelink performed the research and analyzed the data.

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The authors have nothing to report.

Conflicts of Interest

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Parameters used for wb-PBPK model. **Figure S1:** Simulation of ^{18}F -DCFPyL PK in patient 4 as reported in the study by Begum et al. with a simultaneous fit of PET (left) and planar imaging data (right). The solid lines represent predictions generated by our model, while dots correspond to values reported by Begum et al. **Figure S2:** (A–E) ^{18}F -DCFPyL tissue distribution over time in tumor tissue (blue), arterial blood (red), lungs (green), and muscles (purple) in a five patients from the external verification cohort with prostate cancer metastases. The model prediction is represented in the solid line, with corresponding observations in points and the two-fold interval in a ribbon. **Table S2:** Fitted tumor receptor density for all patients which underwent total body PET/CT. **Table S3:** Individual prediction errors (%) for each patient and tissue at 120 min. **Figure S3:** The iterative fitting method increased the accuracy and precision of the model's predictions significantly. The Blant-Altman plot before (left) and after (right) iterative fitting of receptor density. **Table S4:** Parameters of prostate cancer population, used for sensitivity analysis.